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# An artificial neural network approach on crystal plasticity for material modelling in macroscopic simulations

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**Abstract.** Anisotropy plays a significant role in engineering, especially in the field of sheet metal forming. This particular characteristic stems mainly from the crystallographic structure of the metals and the influence of the rolling process, inducing preferred orientations of the grains. In this context, the crystal plasticity theory plays an important role as it accounts for the anisotropic nature of the elastic tensor and the orientation dependencies of the crystallographic deformation mechanisms. Despite the advantages and capabilities, the integration of the crystal plasticity theory in macro simulations is hindered by high computational costs. A novel approach aims to rectify this problem through the application of machine learning. Therefore, this work investigates the machine learning of crystal plasticity simulations, whereby the DAMASK simulation kit package is used both as a benchmark for quality and costs as well as for providing a data basis for the training and testing of the neural networks. A phenomenological material model for an AA5083 aluminium alloy provides the training data for a neural network study, testing different input parameters as well as network setups.

## 1. Introduction

Anisotropy as a property can be found in a multitude of fields of study, ranging from physics, computer imagery to engineering and materials science. The basic definition is that a considered property of a respective material exhibits a directional dependency.

Crystal plasticity as a field of study models the mechanical behaviour of metals on the microscopic scale considering the direction or orientation dependent deformation mechanism in polycrystalline structures. Accordingly, this physics-based approach provides excellent means to model the inherent mechanical anisotropy of metals. [1, 2]

Multi scale simulations offer the possibility to integrate, e.g. crystal plasticity models into macro scale simulations on the component level, and therefore increase the quality and reliability of said models. With the emergence of holistic approaches for the design of products, such as the Integrated Computational Materials Engineering (ICME) [3] or the Materials Genome Initiative [4], multi scale simulations come more and more into focus, for example to develop new materials or to model suitable manufacturing routes.

This work focuses on a potential hybrid approach combining the prevalent concept of machine learning and crystal plasticity for multi scale simulation. The focus lies on the machine learning of crystal plasticity and the capability of neural networks to reproduce its predictions.



## 2. State of the art

Data driven approaches, such as machine learning (ML), are becoming more and more widespread in a variety of research areas, from natural language processing to computer vision, economics and engineering. In the case of material science and computational engineering, two main directions can be distinguished.

On the one hand machine learning is used to analyse and classify the micro structure of materials. Chowdhury et al. applied ML for microstructure recognition using a 2D convolutional neural network (CNN) to distinguish and identify micro structural images with respect to their dendrite structures [9]. They propose an approach that involves using image segmentation techniques to extract micro structure features, followed by applying machine learning algorithms to recognise the micro structure. By comparing the performance of several machine learning algorithms, including support vector machines, random forests and convolutional neural networks, they demonstrate the effectiveness of their approach on various microstructure data sets, which highlights the potential of image-driven machine learning methods for micro structure recognition and their importance in advancing materials science research.

In a similar context, Lubbers et al. again used 2D-CNN to reconstruct micro structural images using unsupervised detection and learning of low-dimensional quantitative, compact descriptions of micro structure [15]. The basic approach is to extract low-dimensional representations of micro structural arrangements and features by CNNs. The authors show that their approach is effective for various micro structures and that these low-dimensional set of features can be used for further tasks, such as microstructure classification and reconstruction.

On the other hand, machine learning models are used as surrogate alternatives to model the material-structure-property relationship that describes the transition from micro structure to component-scale properties. Here, one major focus is the reduction of computational costs [10]. In this field of study, Cecen et al. combined a 3D-CNN with spatial statistics to link microstructure and material properties [8]. The 3D-CNN are used to learn the linkage by analysing the micro structure of materials as 3D images and predicting their macroscopic properties. This approach involves training a model using a large data set of different micro structures labelled with their corresponding properties. Based on that training, the model can be used to predict the properties of an unknown material based on its micro structure. The 3D-CNNs were able to efficiently analyse large data sets of complex micro structures and predict their properties. The applications of this technique lies, for example, in the design of new materials with desired properties. Yuan et al. proposed a machine learning framework, which uses a random forest model fed with data from a viscoplastic self-consistent crystal plasticity model in order to obtain the stress strain curve and the final texture of a given micro structure under load [19]. This approach involves creating a reduced-order model using machine learning algorithms, which are trained on data generated by crystal plasticity simulations. The reduced-order model is then used to predict the macroscopic mechanical behaviour of new materials enabling a straightforward ICME analysis. Ali et al. applied an artificial neural networks (ANN) to achieve similar results, whereby they were able to predict both stress strain curves and texture evolution [5]. They present a novel approach that uses ANNs to predict the mechanical behavior of polycrystalline metals at the microscale level. The authors trained their ANNs with data generated by a crystal plasticity finite element method (CPFEM) model, which is widely used to study the mechanical behaviour of polycrystalline materials. The ANNs were trained to predict the stress and strain responses of polycrystalline metals based on their micro structural features such as grain size, orientation and shape. Beniwal et al. explore the use of deep learning techniques for predictive modelling of the material-structure-property linkage [6]. A 2D-CNN is fed micro structure images of a dual phase material, whereby the stress evolution was related to volume fraction and morphology of the phases. The network was able to predict a yield

stress like quantity, in the context of isotropic elastic plastic mechanics. The absence of human interaction for feature extraction and selection is noteworthy. Pandey and Pokharel were able to predict detailed 3D texture evolution during uniaxial tension for given strain increments using deep ANNs [16]. In their approach, an ANN is trained to predict the deformation of a crystal given its initial configuration and the external loads applied. The ANN is trained on a data set of pre-simulated crystal deformations. The resulting model can accurately predict crystal deformation for unknown initial configurations under uniaxial loading conditions.

### 3. Problem formulation and solution approach

As made clear in the previous section, AI and ML provide novel approaches to material science particularly in the context of modern material development initiatives. Besides these applications, a combination of ML and CP may open up new possibilities in terms of modelling anisotropy, especially in the field of sheet metal forming.

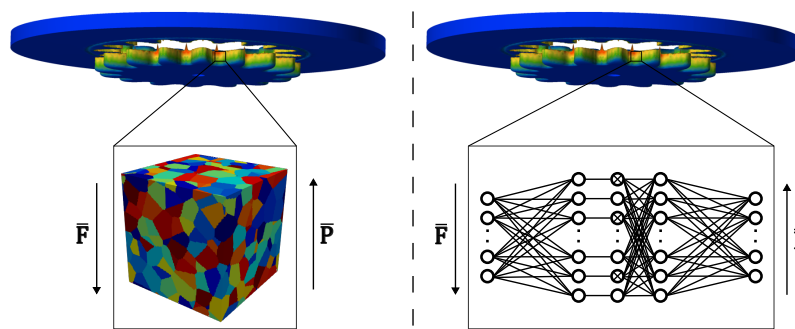
The theory of crystal plasticity is particularly well suited for the simulation of anisotropy, as it takes the tensorial nature of the deformation into account, namely the anisotropy of the elastic stiffness tensor and the orientation dependent activation of the considered deformation mechanisms like twinning or dislocation slip. One drawback of CP simulation on the macro scale is the computational cost connected to it.

The proposed approach aims to reduce computation time of multi scale simulations, similar to surrogate models, whereby the 'on'-time of the simulation is reduced, through computations done in the 'off'-time, accelerating the actual simulation process.

Therefore, the aim of the work is to use machine learning in order to reduce computational costs. In conjunction with this goal is the hypothesis that a neural network trained with suitable features and data can replicate the behaviour of the considered crystal plasticity model.

### 4. Objectives and general approach

The main objective of the present work is to train neural networks with data generated by crystal plasticity simulations in order to replicate the simulated behaviour using the trained network. The successfully trained network is then to replace CP simulations e.g. in multi scale simulations as illustrated by the following figure 1.



**Figure 1.** Integration of NN in multi scale simulations

In comparison to state of the art works as presented in section 2 the proposed approach aims to provide predictions for both texture evolution and mechanical responses, but with a focus on mechanical properties. Furthermore, it is important to emphasise that the present work maps 3D output variables, tensors, and does not resort to averaged and comparative variables, which exceeds the capabilities demonstrated by the state of the art.

The first step of the proposed approach comprises data generation based on a crystal plasticity simulation, preprocessing of the data and a preliminary sensitivity analysis with regards to the characteristics of the representative volume element. Following this, different network architectures are designed and implemented. In the last step the obtained data is again processed and used to train and test the networks capability to correctly predict the mechanical responses of the material under different loads.

## 5. Data Generation

The Düsseldorf Advanced Material Simulation Kit (DAMASK) provides the basis for the data generation [16]. DAMASK is a modular-hierarchical simulation software developed by the Max-Planck Institute für Eisenforschung (MPIE), which has been applied to a number of problems regarding the properties of crystalline materials. The implemented hierarchy is necessary in order to model the hierarchy of scale levels of crystalline materials. The numeric solvers provide the mathematical means to solve the system of partial differential equations, describing the considered boundary problem. Partitioning and homogenisation schemes connect the anisotropic material behaviour on the crystalline level with the global variables. The elastoplastic decomposition in conjunction with elasticity and plasticity sub-modules, i.e. the constitutive laws, allow for the computation of the mechanical response of the material.

Each representative volume element (RVE), which describes e.g. an integration point of a macro scale simulation, encompasses a certain number of constituents or grains with a specific orientation of the basic crystal lattice each in order to represent the material in both the composition in a statistical sense and the material response of the material point with sufficient quality.

The DAMASK software represents this material point model with a material file containing information about the phases, homogenisation, and the applied material model. The geometry file provides the basis for the grid solver, referencing the material in the aforementioned material file. The load file contains information on the applied solver and the loadstep, with details regarding temporal discretisation and the boundary conditions.

### 5.1. Material model

The AA5083 aluminium alloy is chosen for the simulations, characterised by its fcc crystal structure. The phenomenological model used in this case considers dislocations as the only deformation mechanism, as twinning plays a minor role in materials with high stacking fault energies like aluminium and their alloys.

The deformation gradient is multiplicatively decomposed into an elastic and plastic part,

$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_P, \quad (1)$$

$\mathbf{F}_e$  denoting the rotation and elastic distortion of the crystal lattice and  $\mathbf{F}_P$  representing the inelastic shear deformation on the slip planes.

The stress response of the material is computed using the generalised Hook's Law,

$$\mathbf{S} = \mathbb{C} : \mathbf{E}, \quad (2)$$

with  $\mathbf{S}$  being the Second-Piola Kirchhoff stress,  $\mathbf{E}$  being the Green-Lagrange strain and  $\mathbb{C}$  the fourth order anisotropic elastic tensor.

The plastic velocity gradient is composed of the shear contributions of the slip systems  $\alpha$ ,

$$\mathbf{L}_p = \sum_{\alpha=1}^n \dot{\gamma}^\alpha \mathbf{m}^\alpha \otimes \mathbf{n}^\alpha, \quad (3)$$

with  $\dot{\gamma}^\alpha$ , the slip rate of the slip system, the slip direction  $\mathbf{m}^\alpha$  and the normal to the slip plane  $\mathbf{n}^\alpha$ , constituting the Schmidt tensor.

In the context of the chosen material model the shear rates of the slip systems  $\alpha$  are computed as,

$$\dot{\gamma}^\alpha = \dot{\gamma}_0^\alpha \left| \frac{\tau^\alpha}{\xi^\alpha} \right|^n \text{sgn}(\tau^\alpha), \beta = 1, \dots, 12 \quad (4)$$

with  $\dot{\gamma}_0^\alpha$  being the reference slip rate,  $\tau^\alpha$  denoting the resolved shear stress,  $n$  determining the rate sensitivity and  $\xi^\alpha$  being the slip resistance of the respective slip system.

The slip resistance itself evolves asymptotically from an initial value  $\xi_0$  to a saturation value  $\xi_\infty$  following,

$$\dot{\xi}^\alpha = h_0 |\dot{\gamma}^\beta| \left| 1 - \frac{\xi^\beta}{\xi_\infty^\beta} \right|^a \text{sgn} \left( 1 - \frac{\xi^\beta}{\xi_\infty^\beta} \right) h^{\alpha\beta} \text{ with } \beta = 1, \dots, 12, \quad (5)$$

whereby the  $h_0$  is the self hardening coefficient,  $a$  the hardening exponent, and  $h^{\alpha\beta}$  is components of the slip interaction matrix, with a value of 1.0 for coplanar slip systems and 1.4 for non-coplanar slip systems, respectively.

The parameter and constants values are summarised in table 1 and were computed using a Nelder-Mead algorithm.

**Table 1.** Material parameters of AA5083 aluminium alloy.

| $C_{11}$ | $C_{12}$ | $C_{44}$ | $\dot{\gamma}_0$     | $\xi_0$ | $\xi_\infty$ | $h_0$   | $a$  | $n$ |
|----------|----------|----------|----------------------|---------|--------------|---------|------|-----|
| 103 GPa  | 49 GPa   | 53 GPa   | $1.0 \times 10^{-3}$ | 35 MPa  | 138 MPa      | 989 MPa | 2.25 | 20  |

## 5.2. Sensitivity analysis

For the considered case a sensitivity analysis of two critical variables, namely the number of grains per RVE and the spatial resolution was conducted. The main focus was the influence of said variables on the crystal plasticity simulation results.

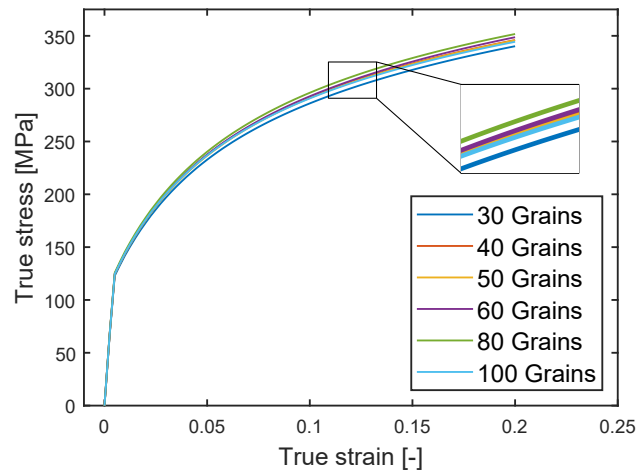
The number of grains has a direct influence on the number of features of the artificial neural networks (cf. table 3), whereas the resolution, i.e. the number of points in the spatial directions of the RVE has a substantial influence on simulation quality, speed and results file size.

In this context, a total number of 28 simulations were carried out, whereby the number of grains is altered in steps from 20 to 100 grains per RVE, with a changing spatial resolution of 16 and 32 grid points in each axis direction. Each single simulations is assigned the same geometry file, but different grain orientations. The considered load case generates uniaxial tension in X-direction, applied as a rate of deformation gradient  $\dot{\mathbf{F}}$  of  $23e - 3$  for 10 seconds.

The obtained results for the true strain and true stress are averaged across three single simulations per alternative, leading to the results as depicted by figure 2. The figure 2 shows the anticipated mechanical behaviour comparable to a basic tensile test, being indicative of the saturation type hardening of the polycrystalline matter.

The results are stable and show no particular sensitivity to the chosen number of grains or resolution. The maximum average difference in stress response is 11 MPa, with a maximum relative difference of 4% with regards to the average stress across all simulations.

The choice of 50 grains per RVE with a spatial resolution of 16 points is therefore well founded, as it is firstly a direct mean to control here reduce, the number of features of the neural networks without compromising the quality of the simulation data and secondly in order



**Figure 2.** Results of sensitivity analysis for uniaxial tension case.

to speed up the data generation and results file size, easing data handling and postprocessing. An increase of features in particular would likely lead to a decrease of information density of the data sets, hampering the training process of the networks.

### 5.3. Simulation setup

Based on the results of the sensitivity analysis, the simulation was set up as follows. For each of the simulations a geometry file with a spatial resolution of 16 points per dimension and a total of 50 grains is used, accordingly with the results of the sensitivity analysis. The chosen resolution of 16 points in conjunction with the chosen number of 50 grains per RVE, results in an average number of  $\approx 80$  grid points or elements per grain, providing a high resolution for each grain. The material file is generated based on the phenomenological material model presented earlier, a total of 50 grains in accordance with the geometry file, whereby the grains orientations are generated randomly. The load cases are set up to generate basic load cases, being composed of uniaxial loading in the three spatial orientations, i.e. tension and compression, and simple shear loading in the the respective spatial orientations, leading to a total of 12 load cases. For the simulations the load was applied defining the rate of deformation gradient  $\dot{\mathbf{F}}$  during the load step and the total time of the load case. For the considered case the spectral grid solver is chosen, due to the advantages regarding computation time in comparison with FEM solvers.

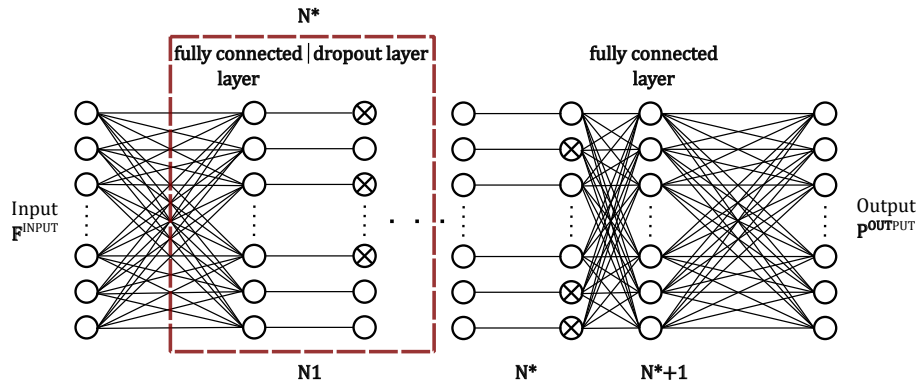
**Table 2.** Simulation parameters.

| Simulation parameter | Value                |
|----------------------|----------------------|
| Solver               | spectral grid solver |
| Grains per RVE       | 50                   |
| Spatial resolution   | 16 x 16 x 16         |
| Orientations         | random               |
| Boundary condition   | $\dot{\mathbf{F}}$   |
| Max deformation      | 25%                  |
| Min deformation      | 1,5%                 |

## 6. Neural Network Architectures

The applied artificial neural networks (ANN) are characterised as multilayer perceptrons (MLPs). A MLP is a feedforward neural network, whereby the input is passed forward through the network. The basic perceptron first described by Rosenblatt, modelled after the structure of human neurons, represent the basic building element of MLPs. The simplest MLP consists of three layers, the input layer, a hidden layer and an output layer. Each layer equals a perceptron, with linear mappings and except for the case of the input function a nonlinear activation function.

Specifically, MLPs with the following structure were used. An input layer supplies the network with the corresponding features, followed by hidden layers consisting of a linear fully connected layer, a nonlinear activation function in the form of the ReLu function and a dropout layer. Finally, the computed results are output. The basic structure of the networks is illustrated in figure 3, whereby the box encloses the hidden layer described above.



**Figure 3.** Multilayerperceptron architecture with dropout layer

In the course of the network study, a number of different network configurations are used. Shallow networks, i.e. a MLP with one hidden layer, are used as those offer sufficient approximation capabilities for many problems. Furthermore, deep networks with 3 and 5 layers each are applied, since deeper networks have proven to be especially useful for more complex problems due to their good approximation capabilities of complex relations.

### 6.1. Data preprocessing

The homogenised deformation gradient in tensor form of the RVE and the initial orientations of the grains as quaternions were defined as features or inputs of the networks. The observations or outputs are the homogenised tensor stress response of the RVE in form of the First-Piola-Kirchhoff stress and the orientation resulting from the deformation. The resulting input and output vectors are concatenated, flattened and have final dimensions of (number of observations x number of features (209)), defining the number of input and output neurons of the neural network architectures.

Due to the difference of the features with respect to their magnitude and the associated risk of exploding/vanishing gradients, a normalisation of the in- and output data is applied before training the ANNs. To prevent data leakage, the entire data set is initially split into training, validation and test data sets following a 60/20/20 split. The total number of files per data set is summarised in table 3 Subsequently, all data sets are normalised to a value range of  $[0,1]$  using the coefficients used on the training data sets.

### 6.2. Training

During training an exhaustive sweep of the parameters in table 4 was conducted. Each combination of hyperparameters was trained using the ADAM optimiser. In contrast to a basic gradient descend algorithm, the ADAM algorithm achieves faster and more stable convergence, using the stochastic first and second moments in order to adapt the basic updating scheme of an ordinary gradient descend implementation.

**Table 3.** Network design specific hyperparameters.

| INPUT variables              | Dimension |
|------------------------------|-----------|
| Deformation gradient(t)      | (3x3)     |
| Initial orientations         | (50x4)    |
| OUTPUT variables             |           |
| First-Piola-Kirchhoff stress | (3x3)     |
| Resulting orientations       | (50x4)    |
| Data sets                    | Size      |
| Training data                | 162.000   |
| Validation data              | 54000     |
| Test data                    | 54000     |

**Table 4.** Network training specific hyperparameters.

| Hyperparameter                                     | Value            |
|----------------------------------------------------|------------------|
| Initial learning rate                              | $1e - 3$         |
| Grad.DecayFactor $\beta_1$                         | 0.9              |
| Squ.Grad.DecayFactor $\beta_2$                     | 0.999            |
| Denominator offset $\epsilon$                      | 1e-8             |
| Mini batch size                                    | [64, 128]        |
| Max Epochs                                         | [30]             |
| Dropout rate                                       | [20%, 40%]       |
| Hidden layers                                      | [1, 3, 5]        |
| Neurons/hidden layer ( $\text{Num}_{\text{neu}}$ ) | [105, 209, 2090] |

The ADAM algorithm in combination with Mini batching is applied for training the networks. Two different initial values for the network learning rate are used in combination with different batch sizes. For the dropout rate a fixed value of 0.4 is set. The weights are initialised according to the approach of Glorot for neural network, whereas the bias values are initialised to zero.

During training the loss, i.e. the squares of the difference between predicted and observed values, and the root mean square error (RMSE), i.e. the square root of the mean of the squares of the difference, are metrics used to asses the quality of the respective networks.

## 7. Results

From the total number of 18 combinations of hyperparameters and network designs the most promising network architecture is chosen based on the RMSE and loss function metrics of the training process. Table 5 shows the chosen configuration and its respective hyperparameter configuration.

In order to provide an engineering metric, the outputs of the test data set are used to compute stress strain curves for specific test cases. The test load cases only represent only a part of the total test data set, but are indicative of the results of the networks.

**Table 5.** Tested network configurations.

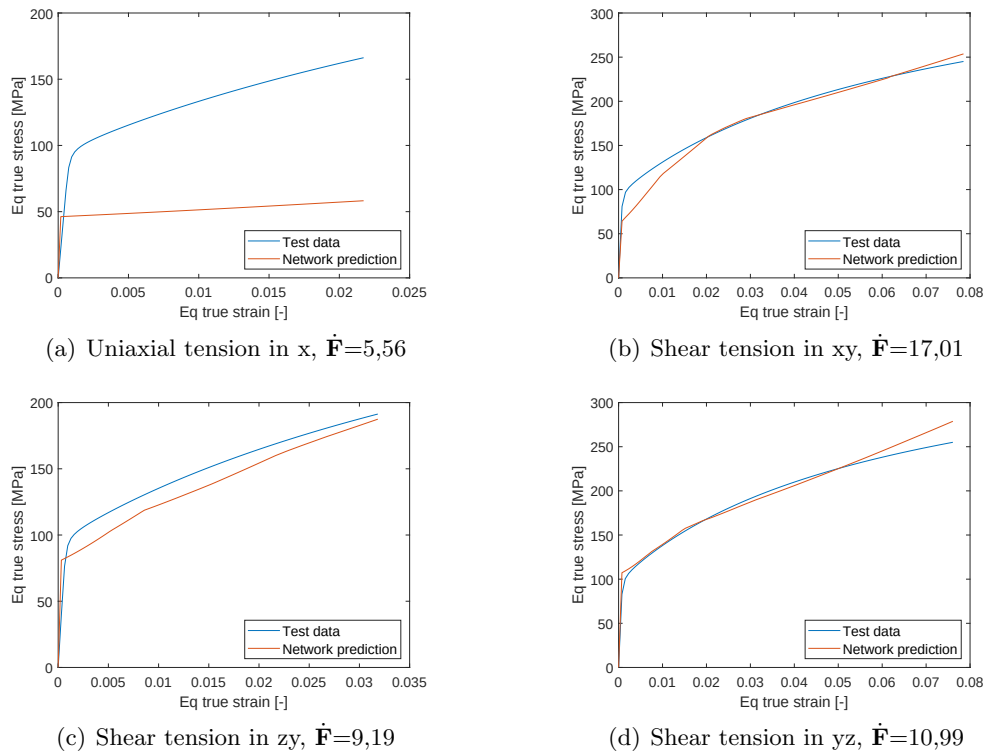
| Hidden layers | $\text{Num}_{\text{neu}}$ | Dropout rate | Minibatch size | Max Epochs | Loss | RMSE  |
|---------------|---------------------------|--------------|----------------|------------|------|-------|
| 1             | 2090                      | 20%          | 128            | 30         | 0.56 | 0.157 |

In order to generate the stress strain curves the output values of the networks are used to compute the true strain  $\bar{\gamma}$ , true stress  $\bar{\sigma}$  and their respective equivalent values using the following relationships,

$$\begin{aligned}\bar{\gamma} &= \frac{1}{2} \ln(\bar{\mathbf{F}}^T \bar{\mathbf{F}}), \quad \bar{\gamma}_{eq} = \sqrt{\frac{2}{3} \boldsymbol{\gamma}^D : \boldsymbol{\gamma}^D} \\ \bar{\sigma} &= \frac{1}{\det(\bar{\mathbf{F}})} \bar{\mathbf{P}} \bar{\mathbf{F}}^T, \quad \bar{\sigma}_{eq} = \sqrt{\frac{3}{2} \boldsymbol{\sigma}^D : \boldsymbol{\sigma}^D},\end{aligned}\tag{6}$$

where  $\boldsymbol{\gamma}^D$  and  $\boldsymbol{\sigma}^D$  represent the deviatoric parts of the tensors.

The network is able to capture the general deformation behaviour, both with respect to the elastoplastic transition and the magnitude of the stress, as depicted in figure 4. The general trend of the stress strain curves, i.e. the saturation type material model used for the crystal plasticity model is captured with good quality. The elastic part of the deformation is modelled correctly with regards to the slope of the graph, though the prediction of the transition from elastic to plastic deformation shows shortcomings of the network. Shear cases are modelled with a high degree of accuracy, both qualitatively and quantitatively. With regards to uniaxial loads the network struggles with modelling the mechanical behaviour. As the uniaxial load cases were split between tension and compression evenly, the information density of the data set was half the density for shear load cases, explaining the deficiencies of the network for those particular load cases.



**Figure 4.** Predicted strain stress curves

## 8. Conclusion

During this study machine learning was applied to the subject of crystal plasticity simulations. The advantages of trained neural networks lie in their adaptability and low computational costs, when compared to sophisticated FEM or FFT simulations. The trained deep NNs are able to predict a full 3D mechanical response with reasonable accuracy.

The considered problem was first analysed with regards to suitable features for the neural networks. A sensitivity analysis showed minimal differences in the obtained simulation results considering different numbers of grains. Subsequently, the chosen phenomenological material model for the AA5083 aluminium alloy was used in conjunction with the DAMASK simulation kit in order to generate both training and test data for the networks. The data was postprocessed to extract the selected features and different network designs with their respective hyperparameters configurations were trained. The networks were subsequently tested and examined with respect to their quality and ability to predict stress responses. The selected features proved to be able to correctly predict the elastoplastic deformation of the RVE using the NNs. The implemented approach is flexible with regards to the types of load cases used and a training of the existing networks with different more complex load cases and or orientations is possible.

Based on the shown performance during the evaluation of the predicted stress strain curves, the stated hypothesis was confirmed. The neural network trained with suitable features and data, can replicate the behaviour of the considered crystal plasticity simulation. Although the networks capabilities were specifically assessed only with regards to the mechanical quantities, the assumption, that the crystallographic quantities are equally well predicted, due to the strong linkage of mechanical outputs and crystallographic variables within the theory of crystal plasticity is reasonable.

Further research could be conducted concerning the application of experimental orientation data gained from EBSD analysis, whereby an orientation density function could be used represent realistic material structures in the CP simulation. Furthermore, the expansion of the repertoire of load cases as indicated earlier could be analysed. Further networks studies centred around the application of different network types, like convolutional neural networks or transformers, to crystal plasticity simulations represent future research topics.

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